

R730xd Backup Node: Build Plan

NetFRAME Home Lab

Drafted 2026-07-22. READ AND FILL THE TBD FIELDS before executing.

1 Purpose and role

A Dell PowerEdge R730xd added as a second, independent storage node whose job is twofold:

1. **Backup target.** It runs its own Proxmox Backup Server (PBS) and *pulls* a copy of Randy's PBS datastore over a scheduled sync job. This gives a real backup of the backups and closes the audit finding that all backups currently sit only on Randy (one node, one rack, one UPS bus). See `audit/findings/30-gap-analysis.md` item R-04.
2. **Transcode node.** It holds the Jellyfin video-transcode GPU, taking that role off Randy.

The sync direction is pull: the backup node holds credentials to read Randy, and Randy stays passive. That is the safer posture (a compromise of the backup node cannot write to Randy).

2 Fields to confirm before you start (TBD)

Field	Value
Node hostname	TBD (pick a name in your scheme)
GPU card	TBD: RX 580 moved from Randy (VAAPI), or a new NVIDIA card (NVENC)
Mgmt IP (VLAN 1)	192.168.10.188 (confirm free in Kea + Pi-hole)
Service IP (VLAN 30, 10G)	192.168.30.188 (confirm free)
iDRAC IP (VLAN 20)	192.168.20.23 (next after Randy .22; confirm free)
Rack location	TBD: a different rack buys real fault isolation for the backup
Sync scope	Randy datastore datastore only (skip the degraded bulk media)

3 Hardware

- Chassis: R730xd, 24x 2.5" front bays (plus rear flexbay).
- Drives: 7x 1.8 TB SAS, 15x 1.2 TB SAS, 2x 250 GB SSD (boot).
- Storage controller: **HBA330 Mini in IT mode** (or a PERC H730 flashed to IT mode), so ZFS owns redundancy. Do not use hardware RAID. This matches Randy's IT-mode approach.
- NIC: a 10 GbE port for the VLAN 30 service path (ConnectX or X520 class), plus the onboard 1 G for VLAN 1 management, mirroring Randy's dual-homed layout.
- GPU: single-slot or dual-slot card in a PCIe riser; confirm the R730xd riser and power support the chosen card (the RX 580 needs a 6-pin/8-pin PCIe power lead).

4 Network (dual-homed, like Randy)

- Management, corosync (if clustered), and the web UIs ride VLAN 1 on the onboard 1 G.

- PBS sync traffic and any NFS ride VLAN 30 on the 10 G NIC, reaching Randy directly at 192.168.30.187 at 10 G. Keep the PBS sync on the .30 path so it never touches the management VLAN.
- iDRAC on VLAN 20 (tagged), reachable from Ares `enp0s31f6.20`, factory creds rotated into Vaultwarden (same policy as the other BMCs).

5 OS and boot

Install Proxmox VE onto the two 250 GB SSDs as a **ZFS mirror** (`rpool1`) using the PVE installer (select the two SSDs, RAID1). PBS installs on top as a service package, the same pattern Randy already runs.

```
# after PVE is installed and updated, add the PBS server package:
apt update && apt install proxmox-backup-server
```

6 ZFS data pool (Option B: 14 active + 1 hot spare)

One pool with three RAIDZ2 vdevs plus a hot spare. Always use `/dev/disk/by-id` names, never `sdX` (which reorder across reboots).

```
# Replace the by-id placeholders with the real device ids (ls -l /dev/disk/by-id | grep -
i sas)
zpool create -o ashift=12 \
  -O compression=zstd -O atime=off -O xattr=sa -O relatime=on \
  vault \
  raidz2 <7x 1.8TB by-id>          '# vdev A: ~8.2 TiB usable' \
  raidz2 <7x 1.2TB by-id>          '# vdev B1: ~5.4 TiB usable' \
  raidz2 <7x 1.2TB by-id>          '# vdev B2: ~5.4 TiB usable' \
  spare <1x 1.2TB by-id>           '# hot spare for the 1.2TB vdevs'

# PBS datastore dataset, larger recordsize suits PBS chunk files:
zfs create -o recordsize=1M vault/pbs
zpool status vault
```

Usable capacity is about 19 TiB, enough to hold Randy's `datastore` PBS data with room to grow. Rename the pool `vault` to your preference if you like.

Spare coverage caveat (important)

The single 1.2 TB hot spare can auto-replace any of the fourteen 1.2 TB drives, but it **cannot** replace a 1.8 TB drive (a spare must be at least as large as the disk it replaces). Vdev A (the 1.8 TB drives) therefore has no automatic spare. Keep a cold 1.8 TB drive on the shelf and replace vdev-A failures by hand.

7 PBS datastore

```
proxmox-backup-manager datastore create backup /vault/pbs
# local retention (independent of Randy) + verify + gc:
proxmox-backup-manager prune-job create backup-prune \
  --store backup --schedule 'daily' \
```

```
--keep-daily 14 --keep-weekly 8 --keep-monthly 6
proxmox-backup-manager verify-job create backup-verify \
  --store backup --schedule 'sun 07:00'
proxmox-backup-manager garbage-collection ... # schedule daily GC in the UI or datastore config
```

8 PBS remote sync (pull from Randy)

On Randy (source): a read-only sync identity

```
proxmox-backup-manager user create sync@pbs
proxmox-backup-manager user generate-token sync@pbs backup-node
# -> copy the returned token secret into Vaultwarden
proxmox-backup-manager acl update /datastore/datastore DatastoreReader \
  --auth-id 'sync@pbs!backup-node'
# note Randy's fingerprint for the remote definition:
proxmox-backup-manager cert info | grep -i fingerprint
```

On the R730xd (target): the remote and the sync job

```
proxmox-backup-manager remote create randy \
  --host 192.168.30.187 \
  --auth-id 'sync@pbs!backup-node' \
  --password '<token-secret-from-vaultwarden>' \
  --fingerprint '<randy-fingerprint>'

# pull Randy's datastore into the local 'backup' datastore, nightly AFTER Randy's jobs finish.
# remove-vanished false: the backup keeps snapshots even after Randy prunes them.
proxmox-backup-manager sync-job create randy-pull \
  --remote randy --remote-store datastore \
  --store backup \
  --schedule '06:00' \
  --remove-vanished false
proxmox-backup-manager sync-job run randy-pull # first run, verify it pulls
```

Randy's nightly guest backups run roughly 02:00 to 08:00 UTC; set the sync schedule to land after they finish (adjust the 06:00 above to your local time as needed).

9 GPU passthrough for Jellyfin

Run Jellyfin in an unprivileged LXC with the GPU passed through. The exact steps depend on the card, which is still TBD.

- **RX 580 (AMD, VAAPI):** bind `/dev/dri` into the container. Jellyfin uses VAAPI for AMD transcode (this is simpler than ROCm, which is not needed for transcode). Ensure the `amdgpu` kernel module loads on the host.
- **New NVIDIA (NVENC):** install the pinned NVIDIA driver on the host plus the container toolkit, then expose the `nvidia` devices to the LXC. This mirrors the QuarkyLab and Jarvis GPU stack pattern (kernel pin plus held driver packages).

Jellyfin media can mount from Randy or from a dataset on this node; decide during setup.

10 Verification checklist

- `zpool status vault` shows all vdevs ONLINE, the spare AVAIL, no errors.
- Reboot once, confirm the pool auto-imports and PVE boots from the SSD mirror.
- `ip -br a` shows the VLAN 30 10 G leg at .30.188, and `ping 192.168.30.187` reaches Randy at 10 G.
- `proxmox-backup-manager sync-job run randy-pull` completes and snapshots appear in the local backup datastore.
- A manual `verify-job` on the local datastore passes.
- Jellyfin transcodes a file using the GPU (check `nvidia-smi` or `radeontop` during a transcode).
- Add the node's node-exporter to Prometheus and confirm the target is UP.

11 Rollback and safety

- This node is purely additive. Nothing on Randy changes except the creation of a read-only `sync@pbs` token, which is revocable: `proxmox-backup-manager user delete 'sync@pbs'`.
- If the sync misbehaves, disable the job (`sync-job update randy-pull --schedule ''`) with no effect on Randy.
- The pool and datastore live only on this node; destroying and rebuilding them touches nothing else.

12 Open follow-ons

- True offsite copy (Backblaze B2 or a remote PBS) as a third layer, later.
- Add this node to the monitoring coverage matrix and alert rules.
- Fill the TBD fields at the top before executing.